Project 3: NorthWest Architecture Studio Network

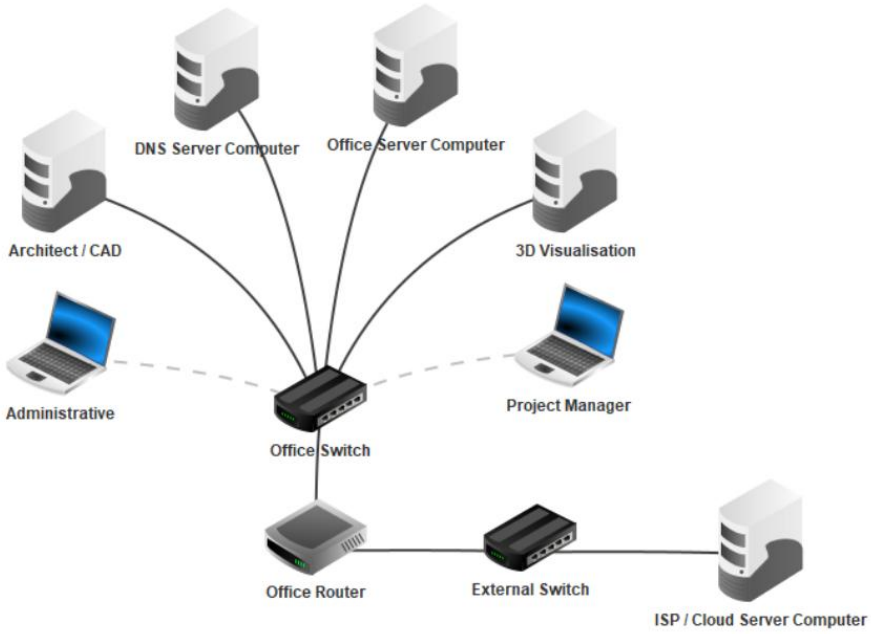
Introduction

In addition to updating its hardware, Northwest Architecture Studio Ltd is also upgrading its IT systems. Staff rely on the network for daily operations. Administrative personnel use the network for sending e-mail and documents; Project Managers utilize the network for sharing reports and planning with other members of their team; and CAD and Visualization staff move large files related to designs across the workstations and servers in the company. Therefore, if the network were poorly constructed, the workflow process would be slowed or even fail due to incorrect routing of information. Therefore, this project will simulate a working office network using Filius and provide the basic concepts necessary to create the simulated office network.

The simulation has two networks. The Office LAN is designated with the address space of 192.168.10.0/24. All equipment (staff PCs, office server, dns server) are connected via an office switch. An Office router connects the Office LAN to an external ISP/cloud network with an address space of 203.0.113.0/24. The Cloud Server resides on the 203.0.113.0/24 network. Each device was assigned a unique IP Address, Subnet Mask and Default Gateway, while the Dns Server resolved names (i.e., "intranet.archstudio.local") into IP Addresses that can be utilized by devices to communicate.

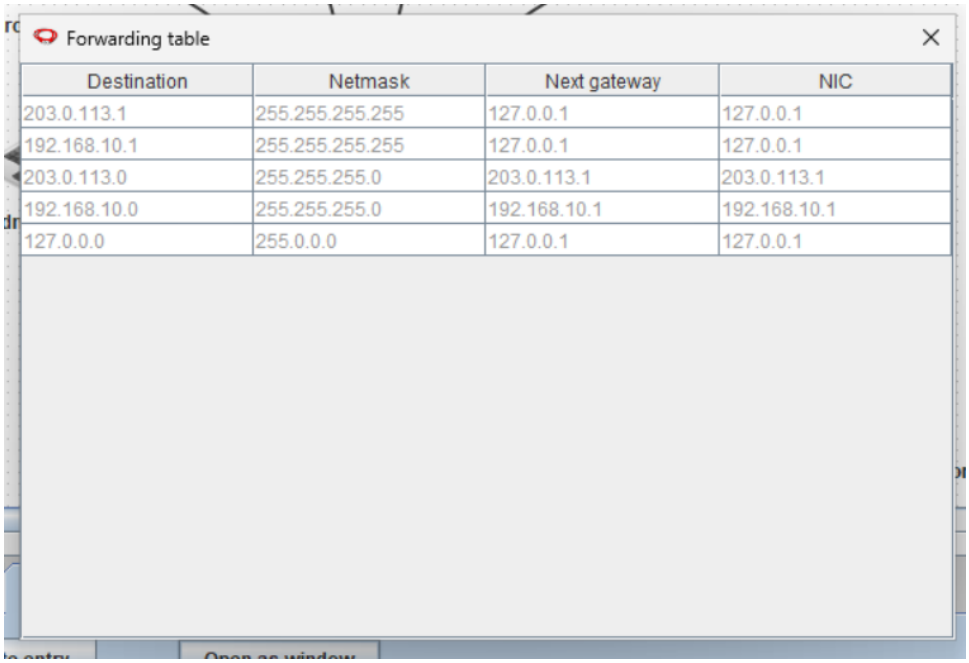
The objective of this phase was to construct the topology, set up addressing and services and lastly, thoroughly test the network from the Architect/CAD workstation. Tests included utilizing Ping to determine connectivity/reachability and Browser Requests over Http to access internal and external web sites both by IP Address and Name. Each Test was selected to illustrate a particular concept. This Section will detail the Evidence collected during testing, explain what each Screenshot illustrates and finally Summarize the Results of Testing.

Topology:



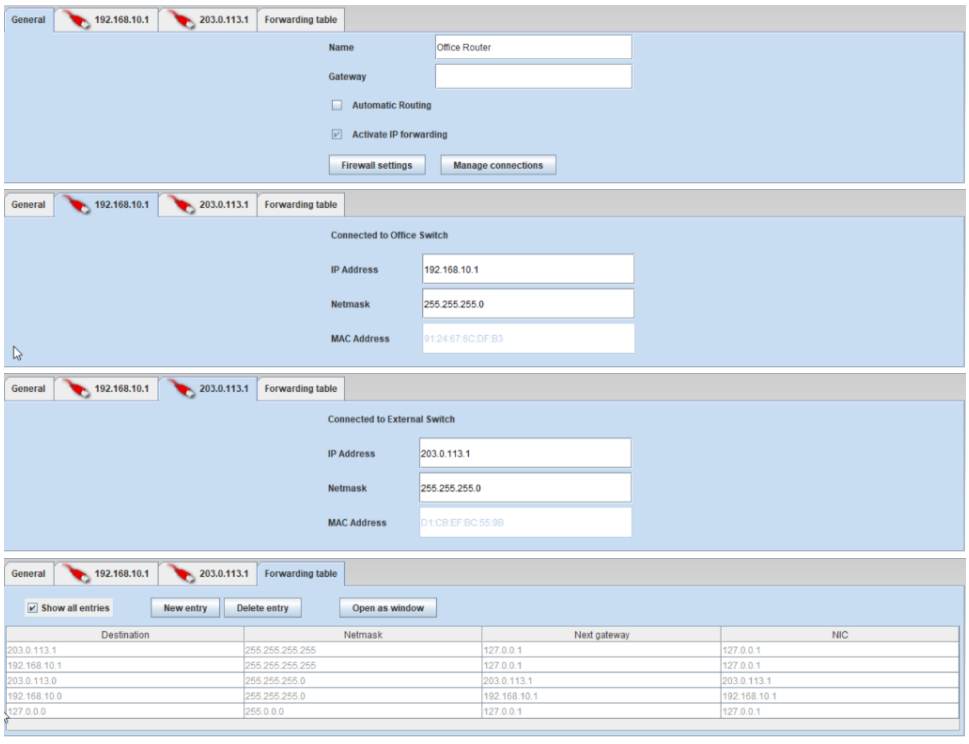
This is a simulation of the entire office network. The four staff computers, the DNS server that resolves web site domain names to IP addresses, and the office server are all connected via the same office switch forming the internal office network. At the edge of the office there is a box referred to as the Office Router that connects the inside office to the outside world represented by an external switch and the Internet Service Provider's cloud server. All data in or out of the office has to travel through this router since it is the only connection point between these networks.

IP Addressing Table:



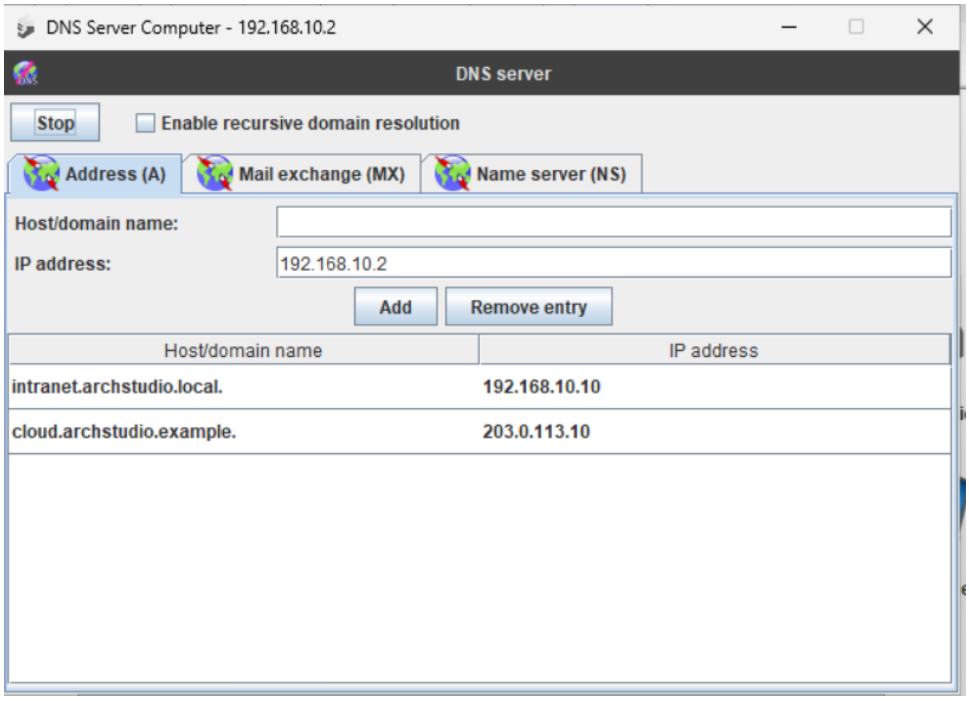
This isn't an address book for the individual devices; this is an internal map the router makes of how to get information from one side of the router to the other. The router is using these rows (there are four total) for its own two connections as well as for the two different networks that connect through it; since the router already "knows" there is another connection on each end, the router can simply transfer a message from one side to the other.

Router-Interface Screenshot:



These screen shots demonstrate the setup of this router. This router is called "Office Router" which is turned ON to provide a link (pass-through) of data between each of its two ports. Port number one is plugged into the office switch. Port #2 is connected to the outside network. Each office computer will send everything that needs to be routed to the outside world to port#1 (192.168.10.1), and then the router will route the message on to port#2 (203.0.113.1). The last picture shows the directory of the router as seen in the previous screen shot.

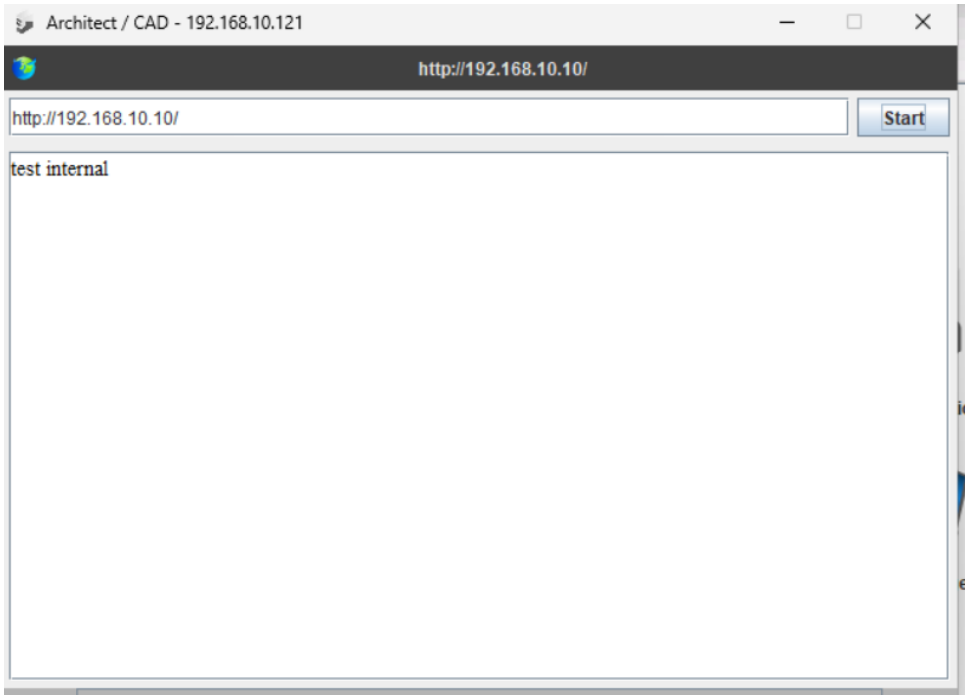
DNS-records screenshot:



This is the DNS server's listing of names. It operates as a telephone directory. You enter in the name and it returns an address. The list contains 2 entries. The first entry is for the company's internal website (intranet.archstudio.local) which refers to the office server at the address 192.168.10.10. The second entry is for the external services (cloud.archstudio.example), which directs users to the cloud server located at 203.0.113.10.

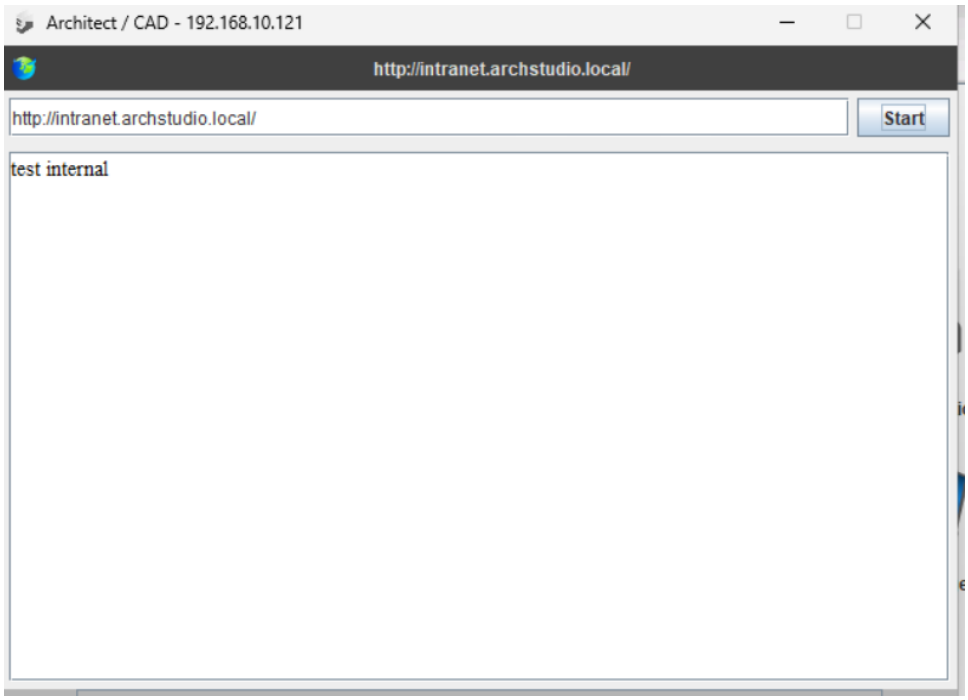
Internal Web Pages (from Architect / CAD Computer):

- Internal Web by IP



Test – opening internal website using address number. When the CAD computer makes an http request to the office server asking it for a specific web page the server returns a page with “Test internal” on it. This shows that computers on the local area network are able to make requests for and get web pages from one another.

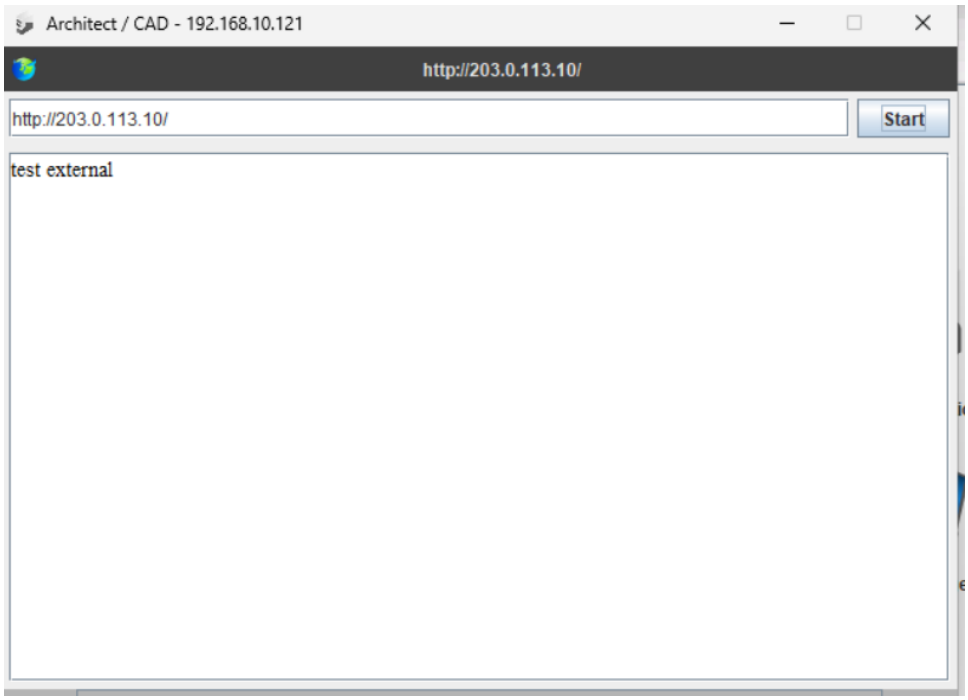
- Internal Web by name



Test – opening same internal website using name. first, the CAD computer checks if there is an entry for the name of the webpage in the DNS phonebook. Once it finds the entry for “internalwebpage”, it uses that to find out where it is located (the ip address) which is 192.168.10.10. Then just like above the browser loads the webpage. The results match what

was seen when the webpage was opened using the address number. Therefore, the name resolution worked properly.

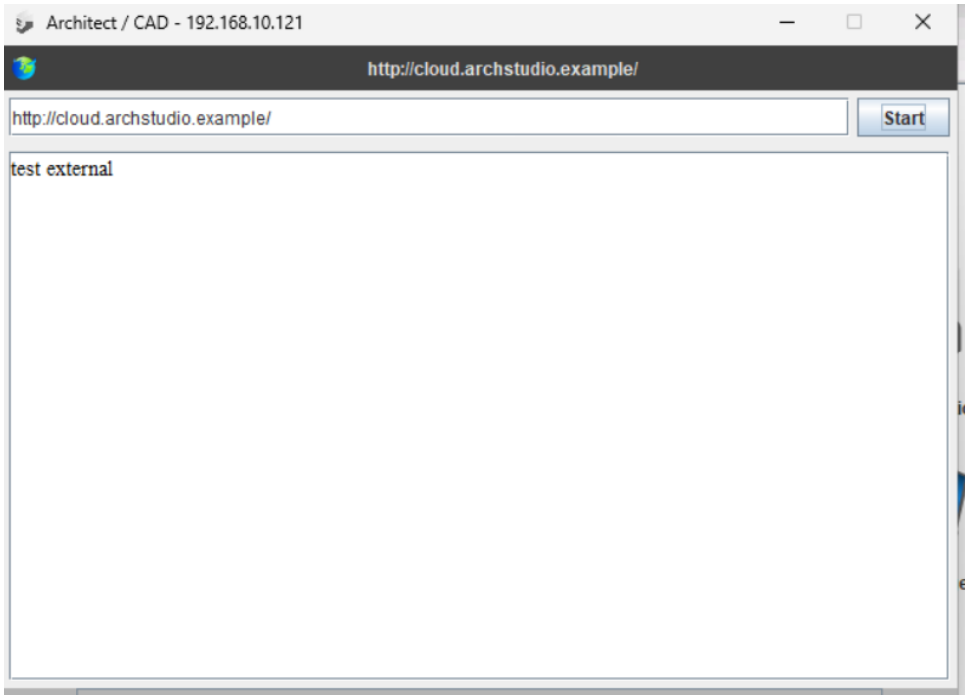
- External Web by IP



TEST: The test opens the outside website using its numeric internet (IP) address. The title of this web-page states “test external”, thus verifying that the requested data was sent to a remote cloud service as opposed to being handled on our internal network.

The process went from the office network to the outside network via the router, which allowed the browser to send requests to an outside domain.

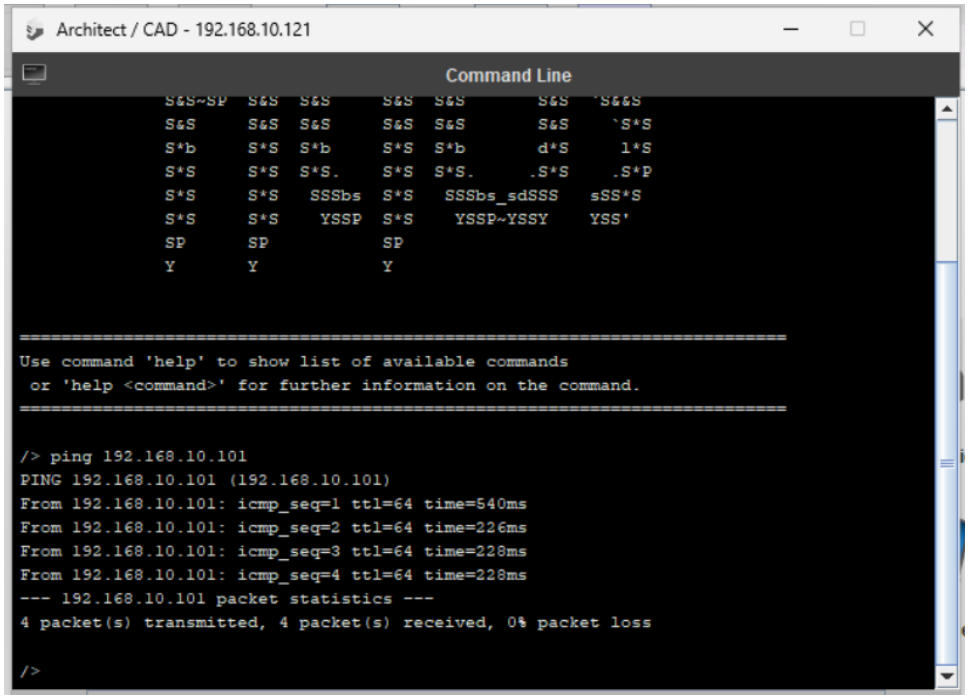
- External Web by name



TEST: The test accesses an outside web-site by name. The name has been translated into an IP Address via a name-lookup (DNS phone-book), with both processes accessing the exact same remote outside web-page.

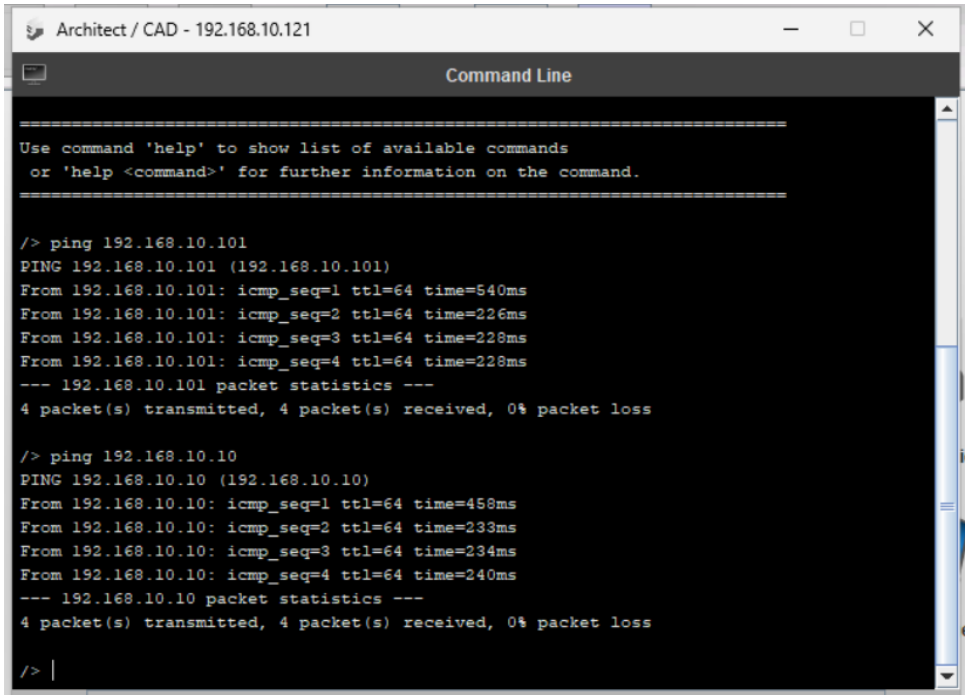
<u>Ping Tests (from Architect / CAD Computer):</u>

- LAN PC-to-PC



Test: “ping” from CAD computer to another office computer. The "ping" command sends an extremely small data package to another computer and then waits for a reply. Since all four replied, this means that these computers are able to communicate with one another by way of the office network (switch).

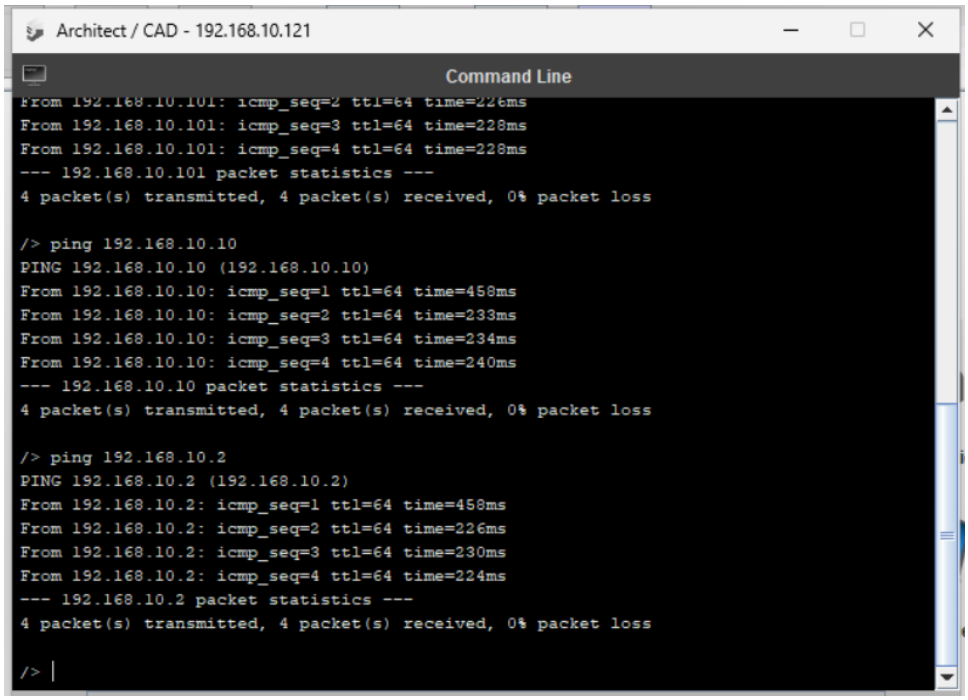
- PC to Office Server



TEST: Ping from CAD Computer to Office Server. The CAD computer was able to send four ping messages to the office server. Each of these messages returned an answer (Reply) from the office server. Therefore the CAD computer has access to the office server. Access to the

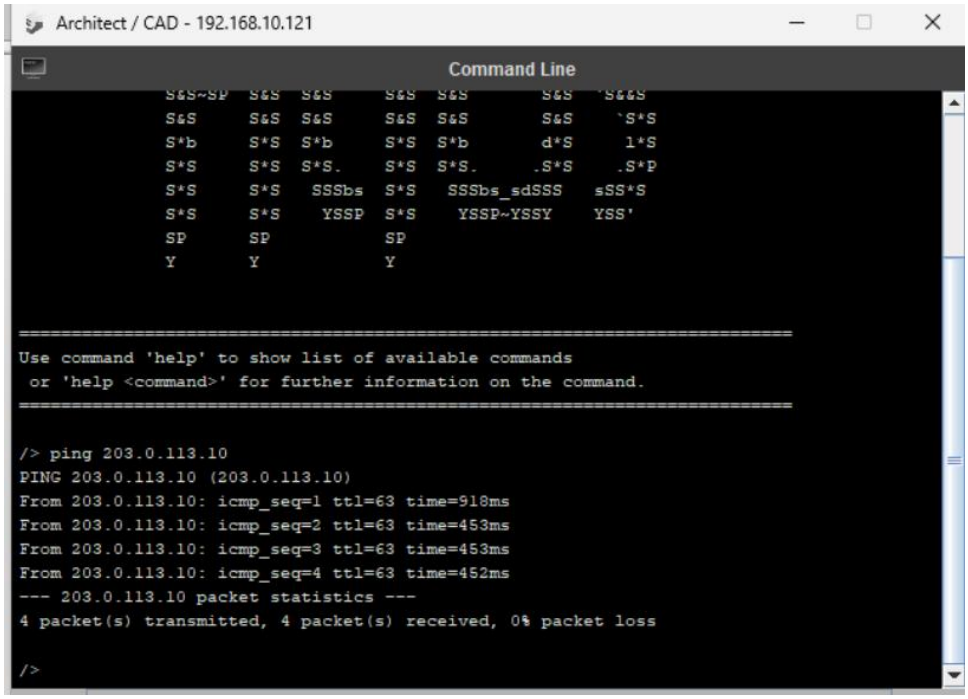
office server is required for this test to be successful as the Internal Website Tests rely on it being available.

- PC to DNS Server



TEST: Ping to the DNS Server. The CAD computer sent four ping messages to the DNS server. Each of these messages returned an answer (Reply) from the DNS server. As such the DNS server appears to be accessible. If it were not then the Name Lookups would fail.

- External Server Ping



Test: a ping (ping means send one packet) to an outside cloud server. All four messages sent out replies. This indicates that the office can access the wider Internet. These reply times were greater than those of the previous test in the office due to increased distance for travel as the packets go through the router and then on to the external switch. As these times are provided by the simulator, they will be different than what you would see with real world internet speeds.

Protocol Table with explanations

Concept	Where it appears	Explanation
TCP/IP	All device communication	Internet Protocol/Transmission Control Protocol: A fundamental framework for communication among computers and/or devices via an internet or network using standard protocols. IP is responsible for routing/addressing by fragmenting large files/messages into packets, with each packet having a specific destination IP address.

		TCP is essentially restoring those packets back to their original file/message, and ensuring accuracy/data integrity by identifying/recovering lost/damaged packets.
IP addressing	Every device has a unique IP	The use of an IP address on a computer is to provide that machine with its own unique number. This will enable the sending of information (data) from the machine in question to be received by the intended recipient machine; also to prevent the sending or receiving of information (data) not intended for either of those machines.
Subnet mask	255.255.255.0 on both networks	A subnet mask divides an IP address into two parts: a Network portion, and a Host portion. Example: 192.168.10.1 - The Network portion would include: 192.168.10 - The Host portion would include: .1
Gateway	Client PCs use 192.168.10.1	A gateway acts as a bridge between networks, allowing data to be transferred through this bridge. For example, between your home network and the internet. It guides data to where it should go, whilst also adhering to security features like firewalls, proxy, encryption, etc.

DNS	intranet.archstudio.local etc.	The Domain Name System (DNS) translates domain names that users find easy to remember, such as "youtube.com" into IP addresses. The reason for this translation is that computers do not understand "youtube.com", but they do understand IP addresses. Therefore, when a user types "youtube.com", their computer sends a request to DNS asking for the IP address associated with the domain name "youtube.com". Once the DNS has translated "youtube.com" into the appropriate IP address, the computer can access YouTube's site and show the requested page.
HTTP	Opening pages in the browser	HTTP is what we use for transferring pages, pictures, etc. from one end of the world to another. A request (browser) is sent to a server and a reply (server) is returned with either the requested webpage or an "error" message.
ICMP / ping	Ping tests	Ping is simply testing whether or not a piece of equipment is accessible. Ping also sends and receives data. When you do a ping it will tell you how long it took for the signal to go through (latency), packets lost (packet loss), and TTL .
Client-server	CAD PC → Office/Cloud Server	A Client-Server Model occurs when you are on your computer asking a server for services/data. The main process of this model is that you ask the server for information, the server processes your request, decides if they will fulfill your request, and then

		responds with either fulfilling your request or denying it.

LAN vs External Network

An A LAN (local area network) is a local area network that can cover one area. For example, your house is a LAN, while your workplace is a LAN. An A LAN allows you to connect various devices in your area including your computer(s), phone(s) and printer(s). A wide area network (WAN) refers to a number of A LAN's that are connected together to provide the ability to retrieve information on servers/data/website across long distances. These distances include around the world.

The process of accessing data or services from a server through a network is called client/server. Your client (device) will send a request to the server asking for some type of data/service. Then the server will decide whether to accept your request and send back either the requested data/service or decline your request based on their policies.

Conclusion

The Network meets all of the Core Requirements outlined within the Brief. Devices on the Office LAN can communicate with each other, the Office Server and the DNS Server; the CAD Computer can access both Internal Web Pages via IP Address and External Web Pages using the Name. All Ping Tests returned successful reply's with 0% Packet Loss; the Router was able to forward Traffic from the Office LAN to the External Network.
Also, the Results demonstrate how the key Concepts Interact. The DNS Record converted a "Name" to an IP Address prior to the Browser making an HTTP Request. Each Client Gateway forwarded Traffic destined for the 203.0.113.x network to the Router, who forwarded this Traffic through his External Interface. Replies from the External Server were received with a TTL of 63 (one less than on the Office LAN), which is consistent with a single Router Hop. Therefore, this verifies that the Router (not either Switch) is the Device connecting the Two Networks.

The Model does have limitations. There are many Components that a Real Firm would require to complete their Network Setup (i.e., Modem/Fibre Connection to ISP, N.A.T., Firewall Rules & Wireless Security). As specified in the Brief, Subnetting and Routing were deliberately kept Simple. For the Firm, some practical Lessons Learned are Documenting Static Addresses of Servers; Providing Internal Services with DNS Records, allowing Staff to Use Names instead of Numbers; and Testing Connectivity in Stages (Local Devices, then Servers, then External Networks) when Faults Occur.

References

Kurose, J.F. and Ross, K.W. (2021) Computer Networking: A Top-Down Approach. 8th edn. Harlow: Pearson.

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